

Taiwan Semiconductor Manufacturing Company

NYSE: TSM

Testudo Stock Pitch Competition - 1st Edition

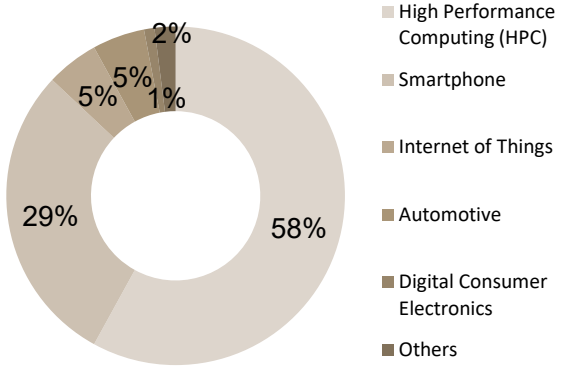
Covadonga Gonzalez-Blas

Company Overview

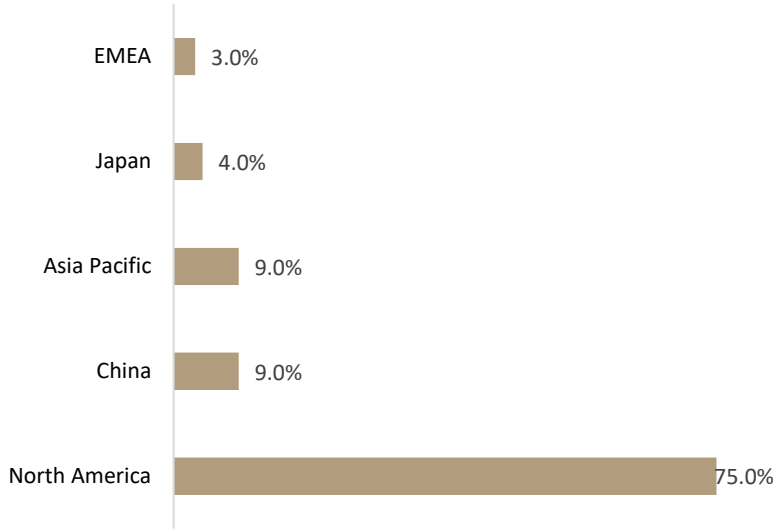
World's largest dedicated chip foundry, the manufacturer behind most leading-edge AI, smartphone and server chips

Snapshot	
Share price	**US$482.3**
Market cap / EV	**US$2.45T / US$2.37T**
Currency	**USD (ADR); NT$ on TWSE**
52-week range	**US$266.8 - 482.3**
TTM revenue (to 2Q26)	**US$142.9bn (+35%)**
TTM operating margin	**56.1%**
Net cash (30-Jun-26)	**US$80bn**
ROE (FY2025)	**35.4%**
Headquarters / founded	**Hsinchu, Taiwan / 1987**

Revenue by platform FY2025



Revenue by location FY2025



Market position

- 72.5% of top-10 foundry revenue in 2Q26
  - c.12x Samsung Foundry's quarterly revenue (US$40.2bn vs US$3.3bn)
- 77% of wafer revenue from 7nm and below; 3nm already 30%, 2nm ramping (3%)

What changed in 2026

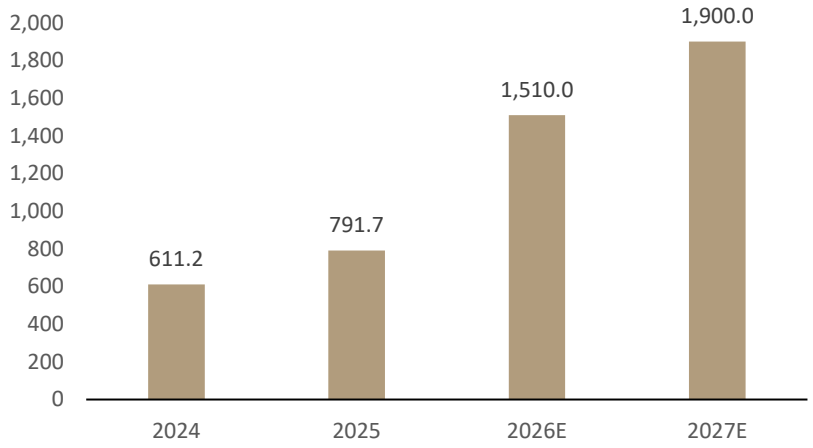
- HPC (AI accelerators, servers) rose to 66% of revenue in 2Q26 from 58% in FY2025; smartphone fell to 22%
- Revenue is guided to grow ~40%+ in US$ in 2026; gross margin hit 67.7% in 2Q26, above its guidance
- Customer geography reflects headquarters of the chip designer, not where end-devices are sold

Sources: Company's filings, Yahoo Finance (price and market cap, 2-Oct-2026) and TrendForce (2Q26 foundry share).

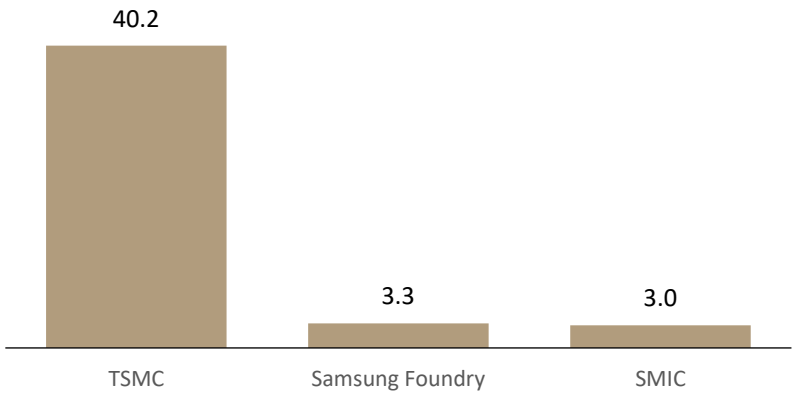
Sector Overview - Semiconductors & Foundry

A US$1.5T industry in 2026E where leading-edge manufacturing capacity, not chip design, is the scarce resource

Total semiconductor market (US$bn)



Foundry revenue, 2Q26 (US$bn)



Market size & growth	- US$792bn in 2025 (+29%) - WSTS: US$1.51T in 2026E and ~US$1.9T in 2027E; logic US$411bn (+37%) then US$523bn (+27%) - H1-26 sales already US$702bn
Structural drivers	AI accelerators and data-centre build-out (TSMC guides AI accelerator revenue CAGR of 54-56% over 2024-29), edge-AI devices, rising silicon content in cars and IoT. Each node costs more, so value concentrates in the few who can build it
Competitive landscape	- TSMC holds 72.5% of top-10 foundry revenue - Samsung Foundry (US$3.3bn) and SMIC (~US$3.0bn) are far smaller; Intel Foundry is still ramping - Moat: process lead, yield learning, CoWoS packaging, co-design with customers
Regulation	- US export controls on advanced chips to China; CHIPS Act incentives tied to US fabs; tariff and Section 232 exposure - China was 9% of FY25 revenue (11% in FY24) - Fabs under way in Arizona, Japan and Germany
Cyclicality	- High historically: FY2023 revenue -8.7% (US$) and ADR -56% peak-to-trough in 2022 - Memory is forecast to grow over 250% in 2026, historically a sign of sector exuberance and potential cyclical risk

Sources: WSTS Autumn-2025 and Spring-2026 forecasts; TrendForce (2Q26 foundry ranking).

Investment Thesis - Three Reasons to Own TSM

A scarce, self-funded, widening-moat asset sitting at the centre of AI compute demand

1 The bottleneck of AI compute

66%

of 2Q26 revenue from HPC (58% in FY25)

- 72.5% of top-10 foundry revenue
- Management guides AI accelerator revenue CAGR 54-56% (2024-29) and ~40%+ revenue growth in 2026
- US revenue doubled in four years: US$56.8bn (FY21) to US$122.4bn (FY25), a 21% CAGR

2 Pricing power turned into cash

67.7%

gross margin in 2Q26 (59.9% FY25, 51.6% FY21)

- Operating margin 50.8% (FY25) to 60.3% (2Q26)
- ROE 35.4% in FY25
- FCF of NT$1.0tn in FY25 after NT$1.27tn capex; NT$3.0tn cumulative FCF FY21-25
- Net cash NT$2.5tn (US$80bn): net debt/EBITDA -0.8x, so growth is self-funded

3 A moat that keeps widening

77%

of wafer revenue from 7nm and below (2nm now ramping)

- 2026 capex guided at US$60-64bn: ~2.5x the annual foundry revenue of Samsung and SMIC combined (~US$25bn)
- 3nm 30% and 2nm 3% of 2Q26 wafer revenue; advanced packaging (CoWoS) is a second capacity constraint it is expanding
- Overseas fabs (US, Japan, Germany) reduce tariff and geopolitical exposure over time

Sources: Company's filings.

Portfolio Fit

Core growth anchor at ~4% of the sleeve: enough to matter, small enough that no single scenario breaks the 30% drawdown limit

Role & sizing

- Role: long-term growth anchor - AI-infrastructure 'picks and shovels' with net cash and 35% ROE
- Weight: 4% of sleeve (range 3-5%, hard cap 5%) vs 2.5% equal weight
- Size: ~GBP 0.8m on an assumed GBP 20m sleeve = 1.6% of the client's GBP 50m
- Dividend ~0.8%: no income need, so total return is all growth

Staged entry (40 / 30 / 30)

- 40% now (after mid-Oct Q3 results)
- 30% on a 10% pullback from the 52-week high (about US$430 or lower)
- 30% once the N2 ramp and 2027 capex plan are confirmed, or after six months

Drawdown stress test (4% weight)

Scenario	TSM move	Sleeve	Total assets
Tariff shock Jan-Apr 25	-36%	-1.4%	-0.6%
Peak-to-trough 2022	-56%	-2.2%	-0.9%
Stress: -40% from today	-40%	-1.6%	-0.6%
Severe: Taiwan / AI bust	-60%	-2.4%	-1.0%

Why it fits a moderate-risk client

- The stock can fall more than 50% (it did in 2022), but at 4% it would take a 125% loss to cost the sleeve 5%
- Worst case modelled uses 3% of the client's 30% discomfort budget
- Pair with non-Asia, non-semiconductor holdings in the other ~39 positions

Main risks & mitigants

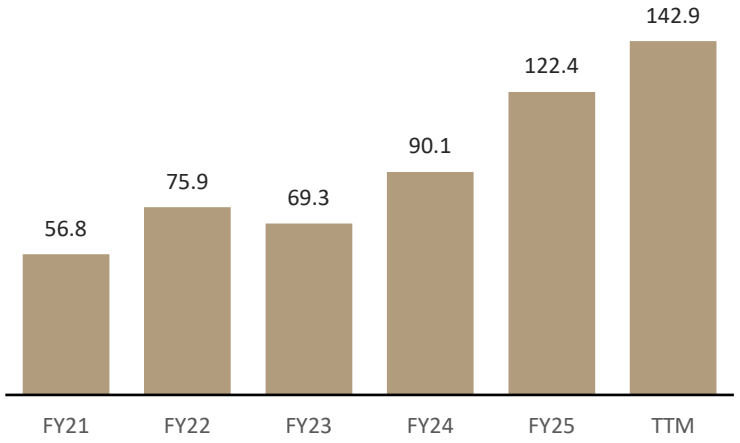
Taiwan / cross-strait	Weight cap; overseas fabs under way
AI capex digestion	Staged entry; Bear case modelled (-54%)
Valuation (TTM P/E 34.5x vs 20.6x 5Y avg)	Tranches; PEG ~1.0x on 2027E
Margin dilution: N2 ramp, overseas fabs	Margin fade to 50% in Base case
Currency: USD ADR vs GBP client	Accept or hedge at sleeve level
Customer concentration (HPC 66%)	Diversified end-markets; ~500 customers

Assumptions: sleeve size GBP 20m, 40 equally weighted positions as benchmark of 2.5%, GBP/USD 1.33.

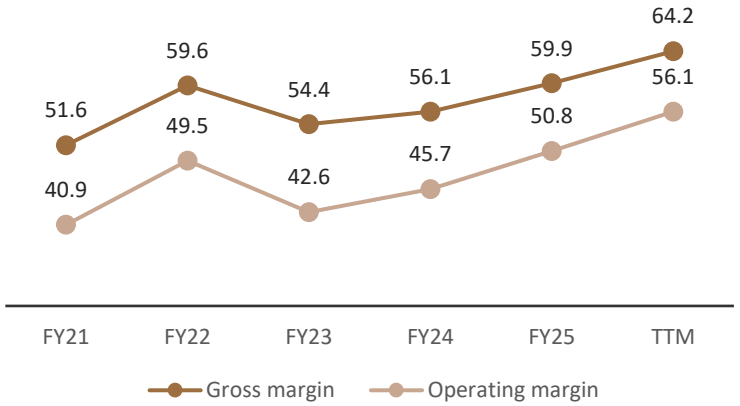
Financials

Five years of 21% revenue CAGR, 46% average operating margin and rising returns - all funded from internal cash

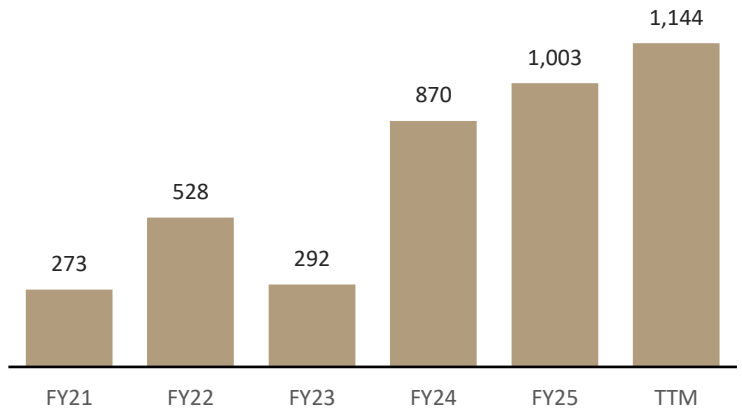
Revenue (US$bn)



Gross and operating margin (%)



Free cash flow (NT$bn)



	FY2021	FY2022	FY2023	FY2024	FY2025	TTM Jun-26[2]
Revenue growth (US$)	+18.9%	+33.5%	-8.7%	+30.0%	+35.9%	+34.9%
Operating margin	40.9%	49.5%	42.6%	45.7%	50.8%	56.1%
FCF margin	17.2%	23.3%	13.5%	30.1%	26.3%	25.8%
Net debt / EBITDA	-0.42x	-0.45x	-0.52x	-0.73x	-0.79x	-0.80x
Return on equity[1]	29.7%	39.8%	26.2%	30.3%	35.4%	n.a.
Diluted EPS (NT$)	23.01	39.20	32.34	45.25	66.25	86.27

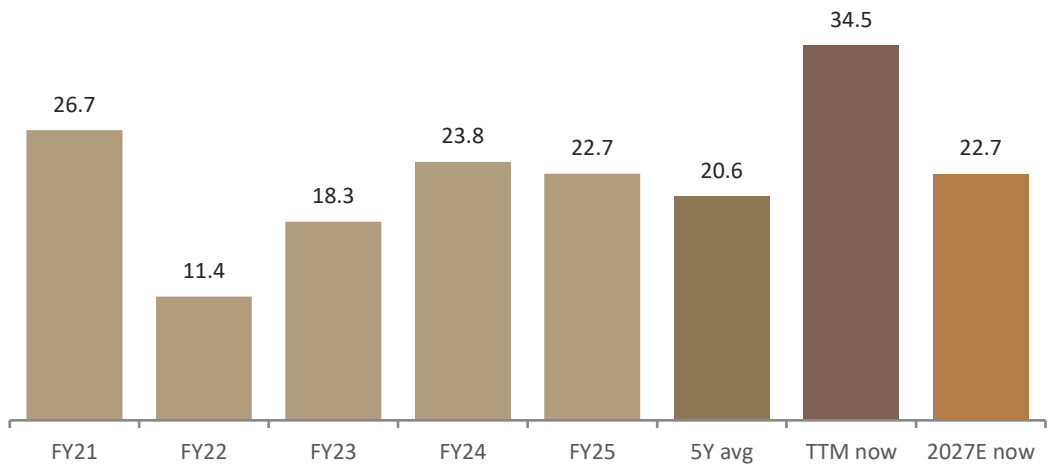
Sources: Company's filings
Notes:
1 ROE as reported by TSMC
2 TTM = four quarters to 30-Jun-2026

Valuation

Expensive on trailing earnings, reasonable on 2027E estimates

	TSM	ASML	Samsung	TSM 5Y avg
Market cap (US$bn)	**2,452**	717	1,250	-
P/E - TTM	**34.5x**	58.9x	13.5x	20.6x
P/E - forward (consensus)	**20.1x**	33.7x	4.4x	-
P/E - 2027E (our Base)	**22.7x**	-	-	-
EV/EBITDA - TTM	**23.5x**	52.7x	7.0x	11.8x
EV/EBITDA - 2027E (Base)	**15.0x**	-	-	-
Operating margin - TTM	**56.1%**	37.1%	36.9%	45.9%

TSM P/E: own history vs. today (x)



US$455

Base DCF value per ADR (-4% vs price)

US$446

Probability-weighted value (25/50/25)

PEG 1.05x

2027E P/E 22.7x / EPS growth 21.7%

~8% p.a.

Base 5-year return to end-2030 (exit 20x)

Verdict: quality compounder at a fair price. Own it at 4% of the sleeve, build in tranches, and add below ~US$430 (10% under the high), where the margin of safety improves.



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Appendix

Testudo Stock Pitch Competition - 1st Edition

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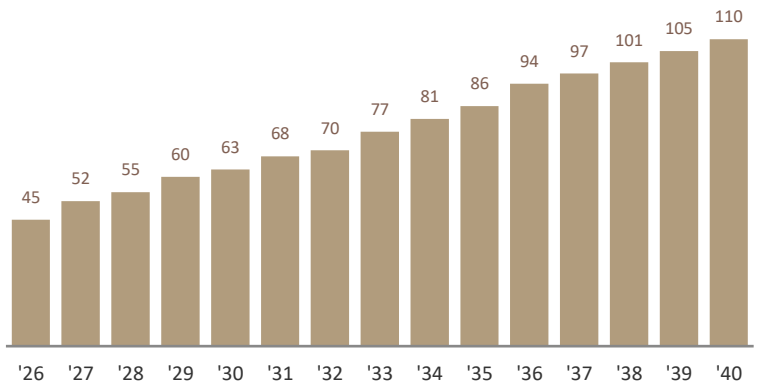
Appendix I – DCF Modeling - Bear Case

Assumptions

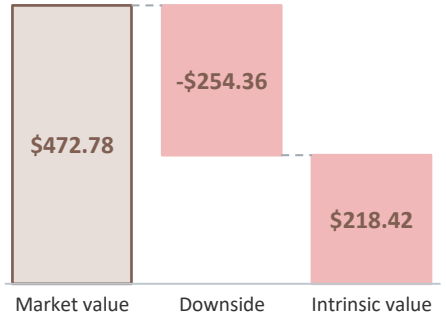
Tax rate	15%
Discount rate (WACC)	10.0%
Perpetual growth rate	3.5%
Risk-free rate (US 10Y)	4.8%
Equity risk premium	4.5%
Adjusted beta	1.17
Valuation date	5-Oct-2026
Current ADR price	$472.78
ADR-equivalents (m)	5,187
Net cash, Dec-25 (US$bn)	65.5

Revenue 2025A (US$bn)	122.4
Rev. growth 26E-30E	38/15/8/6/5%
Rev. growth 31E-35E	5% > 4%
Rev. growth 36E-40E	4% > 4%
Op. margin 26E/30E/40E	57.5/44/42%
Capex % rev 26E/30E/40E	36/28/25%
D&A % rev 26E > 40E	16% > 21%
NWC (% of change in rev.)	8%
Revenue 2035E (CAGR)	$287bn (8.9%)
Explicit period	2026E-2040E

Cash Flow - unlevered FCF (US$bn)



Market vs. Intrinsic Value



Discounted Cash Flow (US$bn)

	2026E	2027E	2028E	2029E	2030E	2031-40E	Terminal
Revenue	168.9	194.3	209.8	222.4	233.5	2,944.0	
EBIT	97.1	101.0	100.7	102.3	102.8	1,238.9	
Less: cash taxes	*14.6*	*15.2*	*15.1*	*15.3*	*15.4*	*185.8*	
Plus: D&A	*27.0*	*32.1*	*35.7*	*38.9*	*42.0*	*598.8*	
Less: capex	*60.8*	*64.1*	*65.0*	*64.5*	*65.4*	*754.3*	
Less: change in NWC	*3.7*	*2.0*	*1.2*	*1.0*	*0.9*	*9.2*	
Unlevered FCF	**45.1**	**51.8**	**55.0**	**60.4**	**63.1**	**888.4**	**1,744.6**
Discount factor	*0.953*	*0.867*	*0.788*	*0.716*	*0.651*	-	*0.251*
PV of FCF	**43.0**	**44.9**	**43.3**	**43.3**	**41.1**	**340.3**	**438.0**

Source: DCF model

WACC x terminal growth (US$ / ADR)

	9%	9.5%	10%	10.5%	11%
2.5%	$236	$220	$206	$193	$183
3%	$245	$227	$212	$198	$187
3.5%	$256	$235	$218	$204	$191
4%	$268	$245	$226	$210	$197
4.5%	$284	$258	$236	$218	$203

WACC x exit EV/EBITDA (US$ / ADR)

	9%	9.5%	10%	10.5%	11%
6x	$209	$201	$193	$186	$179
8x	$234	$224	$215	$206	$198
10x	$258	$247	$237	$227	$218
12x	$283	$270	$258	$247	$237
14x	$308	$293	$280	$267	$256

Terminal Value & Valuation

Terminal value (perpetuity)	1,744.6
PV of terminal value	438.0
Implied exit EV/EBITDA	7.9x
Terminal value % of EV	44%
Sum of PV of FCFF	555.9
Enterprise value	993.9
+ Net cash (Dec-25)	65.5
x Roll-forward to 5-Oct-26	1.069x
Equity value today	1,132.9

Intrinsic value / ADR	**$218.42**
Market price / ADR	**$472.78**
Upside / (downside)	**-53.8%**

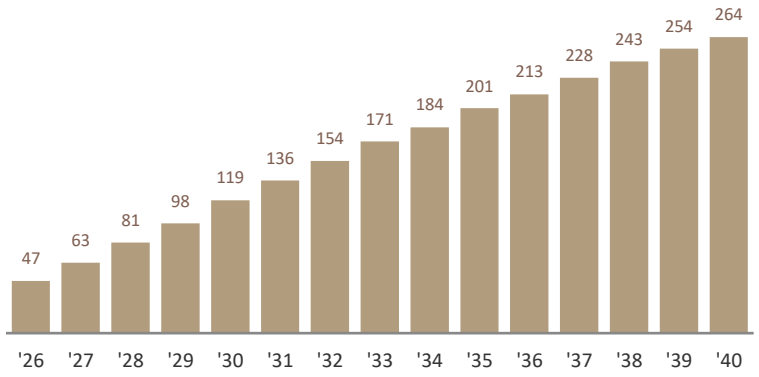
Appendix II – DCF Modeling - Base Case

Assumptions

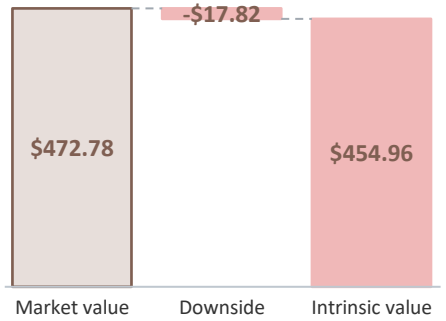
Tax rate	15%
Discount rate (WACC)	10.0%
Perpetual growth rate	3.5%
Risk-free rate (US 10Y)	4.8%
Equity risk premium	4.5%
Adjusted beta	1.17
Valuation date	5-Oct-2026
Current ADR price	$472.78
ADR-equivalents (m)	5,187
Net cash, Dec-25 (US$bn)	65.5

Revenue 2025A (US$bn)	122.4
Rev. growth 26E-30E	41/28/20/16/13%
Rev. growth 31E-35E	11% > 6%
Rev. growth 36E-40E	6% > 4%
Op. margin 26E/30E/40E	58/53/50%
Capex % rev 26E/30E/40E	36/28/23%
D&A % rev 26E > 40E	16% > 21%
NWC (% of change in rev.)	8%
Revenue 2035E (CAGR)	$515bn (15.5%)
Explicit period	2026E-2040E

Cash Flow - unlevered FCF (US$bn)



Market vs. Intrinsic Value



Discounted Cash Flow (US$bn)

	2026E	2027E	2028E	2029E	2030E	2031-40E	Terminal
Revenue	172.6	220.9	265.1	307.6	347.5	5,277.4	
EBIT	100.1	121.5	143.2	164.5	184.2	2,671.0	
Less: cash taxes	*15.0*	*18.2*	*21.5*	*24.7*	*27.6*	*400.7*	
Plus: D&A	*27.6*	*36.5*	*45.1*	*53.8*	*62.6*	*1,076.0*	
Less: capex	*62.1*	*72.9*	*82.2*	*92.3*	*97.3*	*1,274.1*	
Less: change in NWC	*4.0*	*3.9*	*3.5*	*3.4*	*3.2*	*24.8*	
Unlevered FCF	**46.6**	**63.0**	**81.0**	**98.0**	**118.6**	**2,047.4**	**4,206.6**
Discount factor	*0.953*	*0.867*	*0.788*	*0.716*	*0.651*	-	*0.251*
PV of FCF	**44.4**	**54.6**	**63.9**	**70.2**	**77.2**	**774.7**	**1,056.2**

Source: DCF model

WACC x terminal growth (US$ / ADR)

	9%	9.5%	10%	10.5%	11%
2.5%	$497	$458	$424	$395	$369
3%	$518	$475	$438	$407	$379
3.5%	$544	$496	$455	$420	$390
4%	$575	$520	$474	$436	$403
4.5%	$612	$549	$497	$454	$418

WACC x exit EV/EBITDA (US$ / ADR)

	9%	9.5%	10%	10.5%	11%
6x	$409	$392	$375	$360	$345
8x	$462	$441	$421	$403	$386
10x	$514	$490	$468	$446	$426
12x	$567	$539	$514	$489	$467
14x	$619	$589	$560	$533	$507

Terminal Value & Valuation

Terminal value (perpetuity)	4,206.6
PV of terminal value	1,056.2
Implied exit EV/EBITDA	9.0x
Terminal value % of EV	49%
Sum of PV of FCFF	1,085.0
Enterprise value	2,141.2
+ Net cash (Dec-25)	65.5
x Roll-forward to 5-Oct-26	1.069x
Equity value today	2,359.7

Intrinsic value / ADR	**$454.96**
Market price / ADR	**$472.78**
Upside / (downside)	**-3.8%**

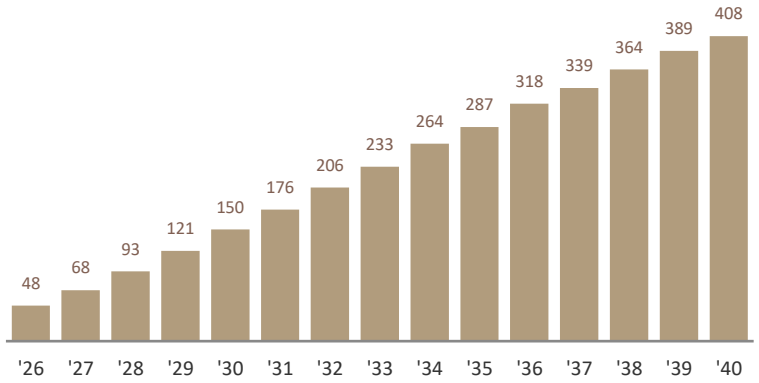
Appendix III – DCF Modeling - Bull Case

Assumptions

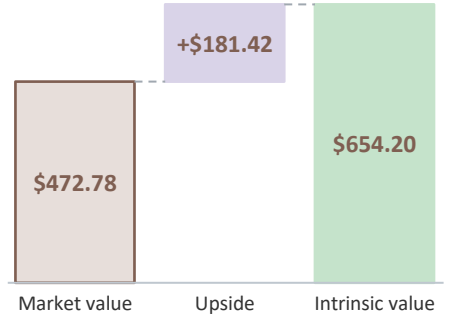
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Rev. growth 31E-35E	14% > 8%
Rev. growth 36E-40E	7% > 5%
Op. margin 26E/30E/40E	58.5/55.5/54%
Capex % rev 26E/30E/40E	36/28/22%
D&A % rev 26E > 40E	16% > 21%
NWC (% of change in rev.)	8%
Revenue 2035E (CAGR)	$685bn (18.8%)
Explicit period	2026E-2040E

Cash Flow - unlevered FCF (US$bn)



Market vs. Intrinsic Value



Discounted Cash Flow (US$bn)

	2026E	2027E	2028E	2029E	2030E	2031-40E	Terminal
Revenue	175.1	236.3	297.8	357.3	414.5	7,036.3	
EBIT	102.4	134.7	168.2	200.1	230.1	3,815.7	
Less: cash taxes	*15.4*	*20.2*	*25.2*	*30.0*	*34.5*	*572.4*	
Plus: D&A	*28.0*	*39.0*	*50.6*	*62.5*	*74.6*	*1,436.7*	
Less: capex	*63.0*	*80.4*	*95.3*	*107.2*	*116.1*	*1,655.9*	
Less: change in NWC	*4.2*	*4.9*	*4.9*	*4.8*	*4.6*	*40.2*	
Unlevered FCF	**47.8**	**68.2**	**93.4**	**120.7**	**149.5**	**2,983.9**	**6,500.6**
Discount factor	*0.953*	*0.867*	*0.788*	*0.716*	*0.651*	-	*0.251*
PV of FCF	**45.6**	**59.2**	**73.6**	**86.4**	**97.4**	**1,113.3**	**1,632.1**

Source: DCF model

WACC x terminal growth (US$ / ADR)

	9%	9.5%	10%	10.5%	11%
2.5%	$717	$658	$607	$562	$523
3%	$751	$685	$629	$580	$538
3.5%	$790	$716	$654	$601	$556
4%	$838	$754	$684	$625	$575
4.5%	$896	$798	$719	$654	$598

WACC x exit EV/EBITDA (US$ / ADR)

	9%	9.5%	10%	10.5%	11%
6x	$571	$546	$521	$499	$477
8x	$649	$618	$589	$562	$537
10x	$726	$690	$657	$626	$597
12x	$803	$763	$725	$690	$656
14x	$880	$835	$793	$753	$716

Terminal Value & Valuation

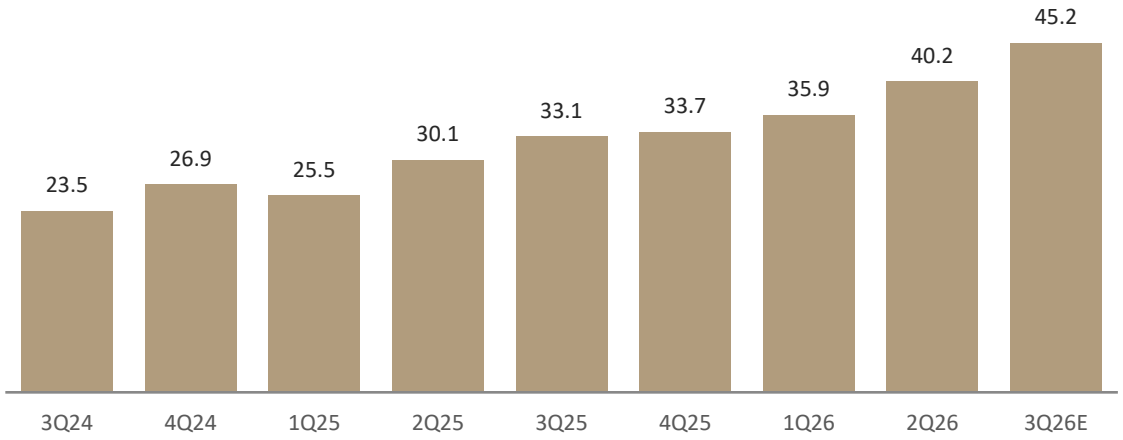
Terminal value (perpetuity)	6,500.6
PV of terminal value	1,632.1
Implied exit EV/EBITDA	9.5x
Terminal value % of EV	53%
Sum of PV of FCFF	1,475.4
Enterprise value	3,107.6
+ Net cash (Dec-25)	65.5
x Roll-forward to 5-Oct-26	1.069x
Equity value today	3,393.1

Intrinsic value / ADR	**$654.20**
Market price / ADR	**$472.78**
Upside / (downside)	**+38.4%**

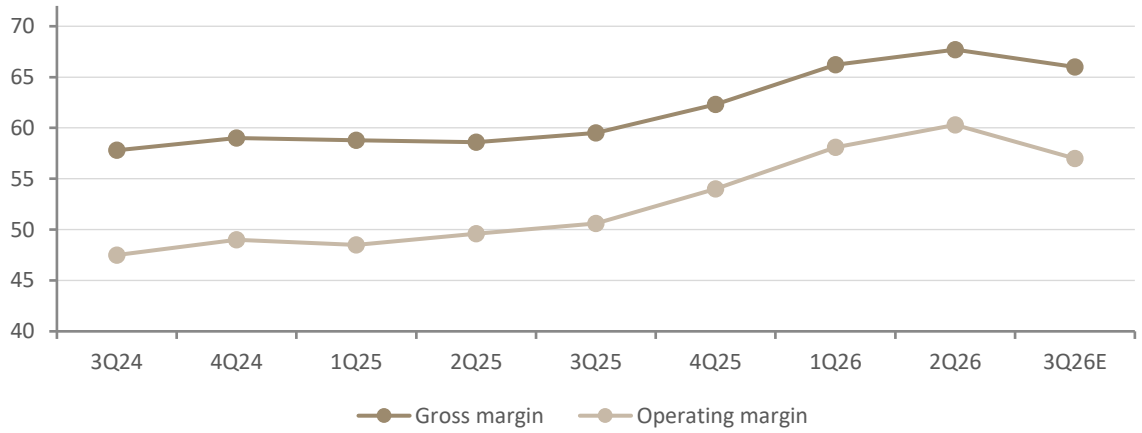
Appendix IV - Quarterly Momentum & Technology Mix

Gross margin is up 9 points year on year to 67.7% while advanced nodes carry over three-quarters of wafer revenue

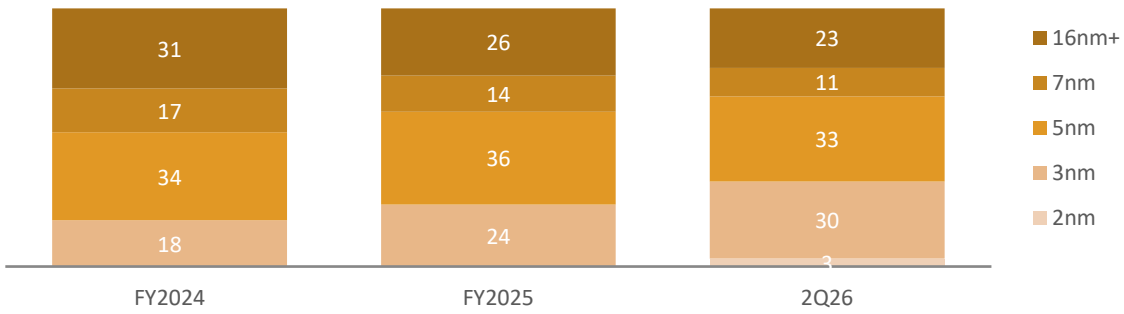
Quarterly revenue (US$bn)



Gross and operating margin (%)



Wafer revenue by process node (%)



What to watch next

- Q3 results in mid-October: US$44.6-45.8bn revenue and 65-67% gross margin guided
- 2026 capex US$60-64bn (raised from US$52-56bn): demand signal, but also a cash-flow drag
- Taiwan policy, US tariffs and export controls; sustained AI customer capex